DC FUND INTELLIGENCE

DC Screening Report

Asset Carbon Risk and Regulatory Exposure Assessment

FACILITY

Clonshaugh DC (Fictional Reference Facility)

DATE

April 2026

PREPARED BY

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CLIENT

Sample Client — For Demonstration Only

REFERENCE

LBE-SCR-CLN-001

CLASSIFICATION

Sample — Demonstration Only

Executive Summary

OVERALL ASSESSMENT VERDICT

HIGH EXPOSURE — Immediate Action Required on Carbon Trajectory and EU Taxonomy

Clonshaugh DC presents significant regulatory and valuation exposure across four of eight assessed criteria. Carbon intensity of 336 $kgCO_2$/MWh_IT places the facility in the misaligned band. PUE 1.50 fails the EU Taxonomy 1.3 threshold. No qualifying renewable procurement is in place. A retrofit programme is technically achievable and economically justified.

336	1.50	278	1.30
CARBON INTENSITY $KGCO_2$/MWH_IT	CURRENT PUE	TARGET CI $KGCO_2$/MWH_IT	TARGET PUE
▲ Misaligned	▲ Fails 1.3 threshold	▲ Below 300 band	✓ Taxonomy pass

REGULATORY EXPOSURE SUMMARY

Eight regulations assessed. RAG per criterion. Green: no material gap. Amber: manageable gap. Red: hard exposure requiring action.

REGULATION	CRITERION	STATUS	KEY FINDING
EU Taxonomy Delegated Act 2021/2139	PUE ≤ 1.3	RED	PUE 1.50 — fails threshold. Retrofit required for taxonomy alignment.
CRU/2025236	80% qualifying renewable	RED	0% qualifying procurement. Obligation triggered on MIC application. Six-year glide path available.
CRREM Pathway (LBE-derived)	Carbon intensity trajectory	RED	336 $kgCO_2$/MWh_IT. Currently misaligned. Misalignment Year approx. 2033 at current trajectory.
EED Article 12	Energy audit obligation	AMBER	2.4 MW IT load — above 500 kW threshold. Audit obligation applies. Confirm last audit date.
EED Article 26	Waste heat reporting	AMBER	Total facility load above 1 MW rated input. Reporting obligation triggered. Assess waste heat potential.
F-Gas Regulation 2024/573	Refrigerant phase-down	AMBER	CRAH plant uses HFC refrigerants. Phase-down timeline requires replacement planning.
Carbon Tax (Finance Act)	Generator fuel exposure	AMBER	€71/tCO_2 rising to €100 by 2030. Diesel generation cost trajectory increases OpEx baseline.
CRM T-4 (SEMO PCAR2829T-4)	Capacity market participation	GREEN	No mandatory participation obligation at current scale. Monitor for threshold changes.

Carbon Position and Retrofit Economics

FACILITY PROFILE

Client-supplied. Used as inputs to all calculations.

Facility	Clonshaugh DC (Fictional)
Year Built	2013
IT Load	2.4 MW (6 kW/rack avg)
Rack Count	400 racks
Current PUE	1.50 FAILS 1.3
Cooling	Raised-floor CRAH (HFC)
Grid	10 kV ESB Networks MV
MIC	5 MVA
Generator	Diesel, 200 hrs/yr
Qualifying PPAs	None

CARBON INTENSITY

Grid EF 0.2241 $kgCO_2/kWh$ (SEAI 2026, T1). Location-based Scope 2.

Current CI	0.2241 × 1.50 × 1,000 = **336.2 $kgCO_2$/MWh_IT**
Target CI (PUE 1.30)	0.2241 × 1.30 × 1,000 = **291.3 $kgCO_2$/MWh_IT**
CI reduction	44.9 $kgCO_2$/MWh_IT (13.3%)

CRREM Disclosure: Data centre pathway bands (200/300/400 $kgCO_2$/MWh_IT) are LBE-derived at T3/T4. The CRREM Foundation has not published an official data centre decarbonisation pathway as at April 2026. Misalignment Year is an LBE metric.

10-YEAR HOLD MODEL

AACE Class 5 (−20% to +50%). Indicative only. Not a design specification.

PARAMETER	BASE	STRESSED
Annual energy saving (MWh)	4,205	4,205
Electricity price (euros/kWh)	0.12	0.15
Annual cost saving	**504,600**	**630,750**
Carbon saving (tCO_2/yr)	942	942
Carbon tax saving (euros/yr)	66,882	94,200
Indicative retrofit CAPEX	400K – 900K	400K – 900K
Simple payback (years)	**< 2**	**< 1.5**

RETROFIT PATHWAY OPTIONS

Option 1 — Cooling Optimisation (Primary Path)
Cold aisle containment, VSD retrofit, free cooling controls upgrade. Achieves PUE 1.30–1.35. CAPEX indicative 400K–900K. Compatible with existing infrastructure. Recommended as first-stage action.

Option 2 — Renewable PPA Procurement
Corporate PPA for qualifying Irish renewable generation. Addresses CRU/2025236 80% obligation triggered on any future MIC application. Six-year glide path from connection date. PPA development timelines 2–4 years — procurement planning should start now.

Option 3 — HVO Generator Fuel Switch
Replace diesel with HVO for backup generation. Reduces Scope 1 emissions. Carbon tax exposure reduced. No structural change to infrastructure. Short lead time.

Methodology and Professional Sign-Off

All outputs are screening-level and indicative — not exact or 100% accurate. LBE does not design, specify, or select equipment or contractors. This is the source of its independence. Because LBE has no interest in recommending a specific contractor or technology, this report is the only truly independent intelligence you can put in front of your investment committee.

This assessment relies on facility data supplied by the client. LBE has not independently verified this data. Where data was unavailable, conservative assumptions have been applied. The report does not substitute for a detailed site survey by the appointed engineering firm. LBE carries PI insurance appropriate to its scope as an engineering intelligence consultancy. LBE does not carry PI insurance for engineering design or delivery.

CRREM disclosure: Data centre pathway bands (200, 300, 400 $kgCO_2/MWh_IT$) are LBE-derived at T3/T4. The CRREM Foundation has not published an official data centre decarbonisation pathway as at April 2026. All references use the LBE-defined Misalignment Year metric. Results should not be attributed to the CRREM Foundation.

CANONICAL VALUES

PARAMETER	VALUE	TIER
Grid EF	0.2241 $kgCO_2/kWh$ (SEAI 2026)	T1
Carbon tax	€71/tCO_2 → €100 by 2030	T1
Electricity	€0.12/kWh (CRU Q4 2024)	T2
EU Taxonomy PUE	≤ 1.3 (Delegated Act 2021/2139)	T1
CRU renewable	80% (CRU/2025236)	T1
CRREM bands	200/300/400 $kgCO_2/MWh_IT$ (LBE T3/T4)	T3
Cost benchmarks	SCSI M&E, RICS NRM1, mfr data	T2/T3

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PROFESSIONAL SIGN-OFF

PREPARED AND SIGNED BY	DATE	STATUS
Les Murphy CEng MBA MSc MIEI Founder, Legacy Business Engineers Ltd | CRO 720301	April 2026	SAMPLE

Standards: EU Taxonomy Delegated Act 2021/2139; SEAI 2026; CRU/2025236; CRREM Foundation (LBE-derived DC pathway); EED Articles 12 & 26; F-Gas Regulation 2024/573; Finance Act (carbon tax); SEMO PCAR2829T-4; AACE Class 5; RICS NRM1; SCSI M&E Benchmarks 2025. Tool: DC-RPT-SCR-001 v1.0.0.